CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

A new diagonal separation and its relations with the Hausdorff property

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Abstract

Let P be a property of subobjects relevant in a category C. An object X in C is P-separated if the diagonal in $X \times X$ has P; thus e.g. closedness in the category of topological spaces (resp. locales) induces the Hausdorff (resp. strong Hausdorff) axiom.

In this talk we consider the locales in which the diagonal is fitted (i.e., an intersection of open sublocales - we speak about F-separated locales). Among others, we will show that F-separatedness is a property strictly weaker than fitness. Moreover, we will explore a pleasant parallel with the strong Hausdorff axiom, including a Dowker-Strauss type theorem and a characterization in terms of certain relaxed morphisms.

This is a joint work with Jorge Picado and Aleš Pultr.